Blackmon Lab

Evolutionary Biology

The Influence of Arboreal and Terrestrial Environment on the r/K Tradeoff in Marsupials

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Background

Life history theory is an evolutionary foundation that predicts a fundamental tradeoff between reproductive output and body size, r/K selection, that explains the idea that species can be small and invest in either many offspring (r) or large and invest in few offspring (K). While this tradeoff is established across mammals, how specific ecological factors such as lifestyle affect this relationship is not well understood. By comparing arboreal and terrestrial marsupial species using phylogenetic comparative methods we can isolate the effect of lifestyle while controlling for shared evolutionary history. Using body mass and litter size data from 159 marsupial species compiled from the PanTHERIA and EltonTraits databases, I tested whether lifestyle influences how the r/K tradeoff manifests across marsupials using PGLS. **I hypothesize that arboreal and terrestrial will both have a negative relationship between body mass and litter size but the strength of the relationship will differ.**

Study Design & AI

The use of AI was an important part of my research in helping with initial planning of my project and generating R code. R code was created for data extraction from PanTHERIA and EltonTraits to get body mass, litter size, and lifestyle data, as well as from TimeTree to make a species list from the phylogenetic tree. Code was used to create figures that displayed the data in a clear way and also to run the PGLS analysis.

Marsupial Lifestyles

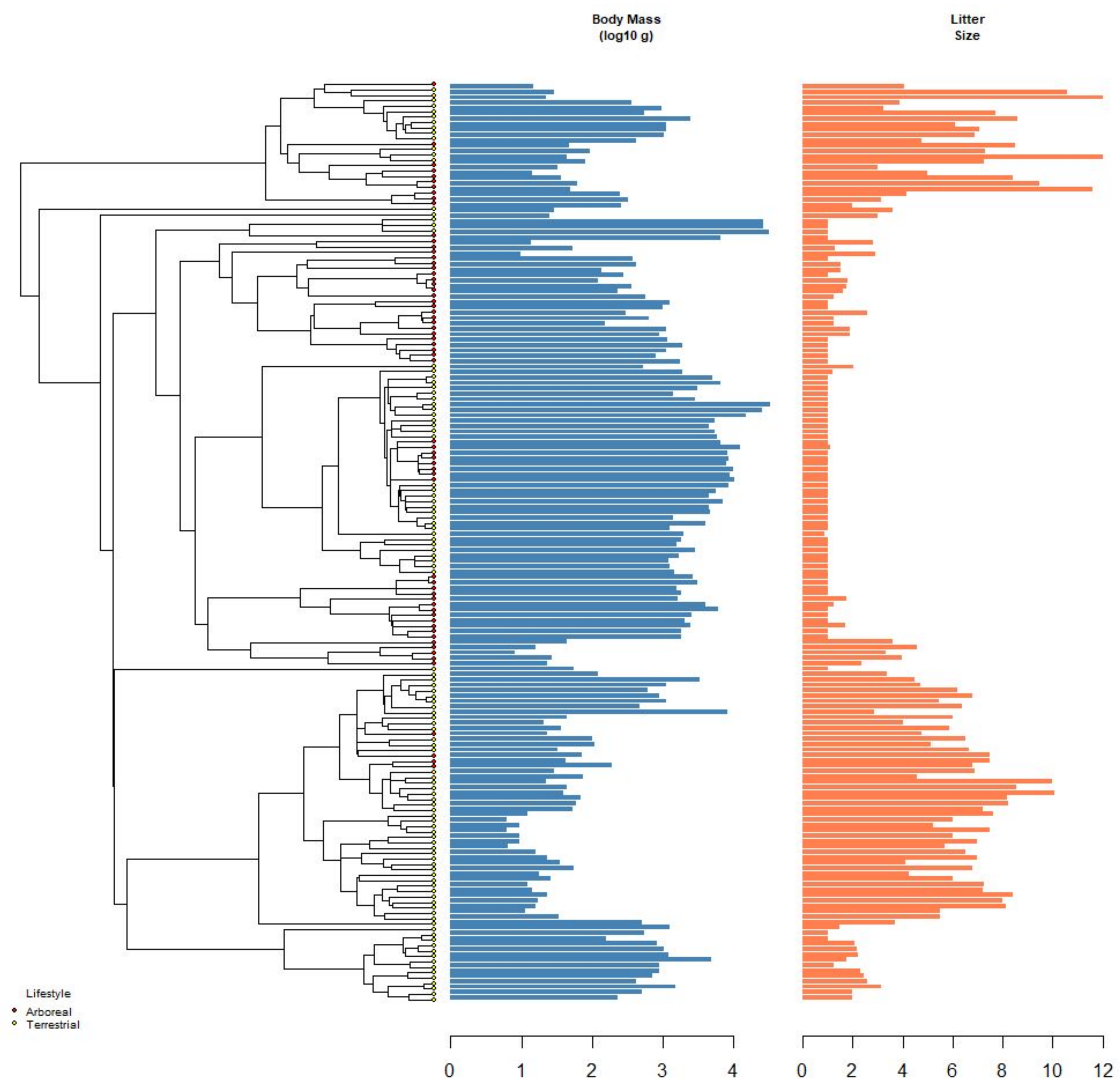
Planigale ingrami (Long Tailed Planigale)
Average body mass of 6.27g and average litter size of 7.48, represents r-selected strategy.

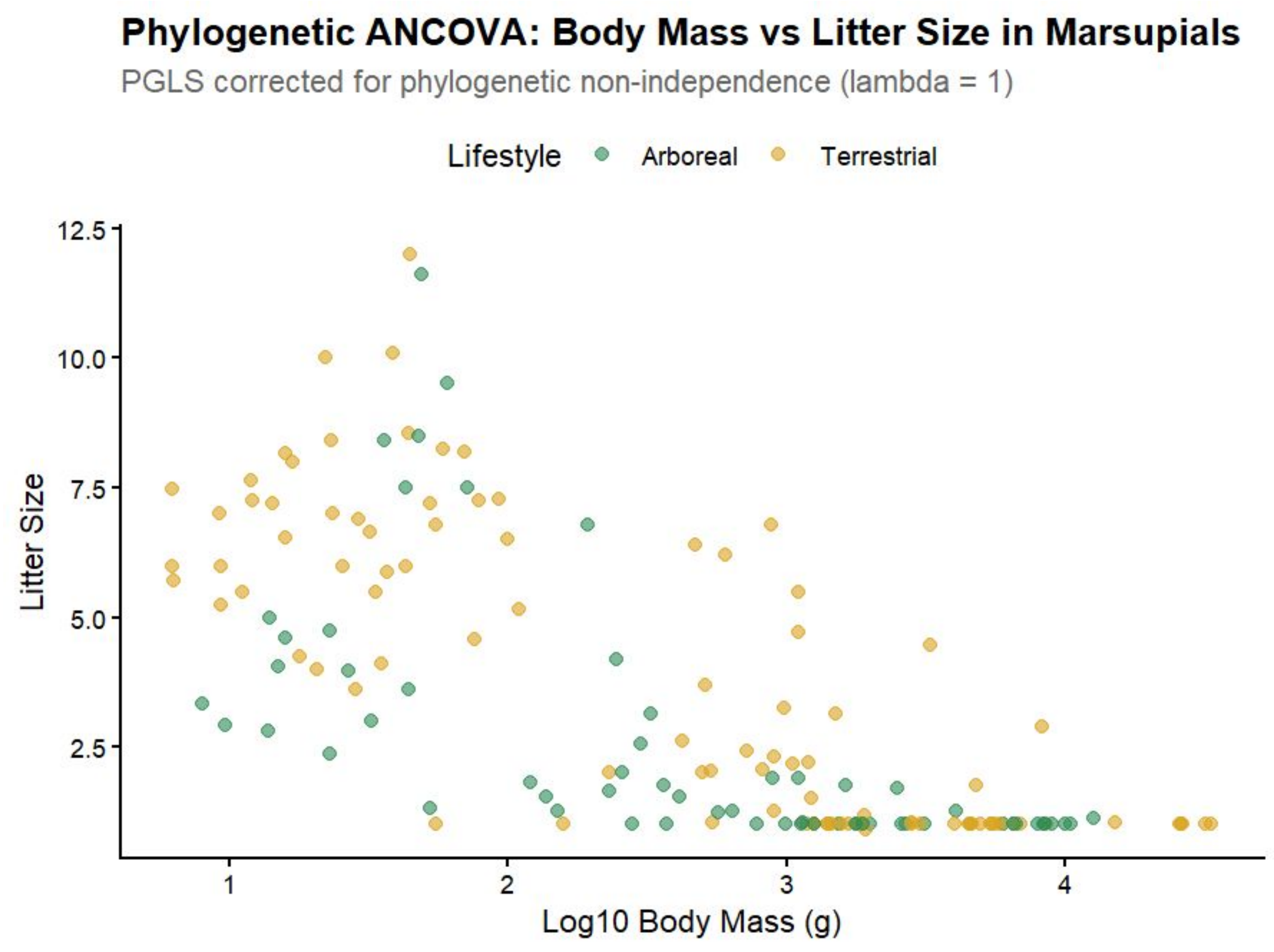
Macropus Giganteus (Eastern grey Kangaroo)
Average body mass of 33409.89g and average litter size of 1, represents K-selected strategy.

Cercartetus Lepidus (Tasmanian Pygmy Possum)
Average body mass of 8.03g and average litter size of 3.33, represents r-selected strategy

Phascolarctos cinereus (Koala)
Average body mass of 6528.74g and average litter size 1.01, represents K-selected strategy.

Results





Term	Estimate	StdErr	t.value	p.value
Baseline Litter Size (Arboreal)	1.8821	1.3067	1.4404	0.1518
Effect of Body Mass (Arboreal)	0.5631	0.0024	234.0576	< 2.2e-16 ***
Terrestrial vs Arboreal (Baseline)	2.8798	0.8566	3.3620	0.0010 ***
Difference in r/K Slope (Terrestrial vs Arboreal)	-1.0045	0.2560	-3.9245	0.0001 ***

Discussion

I hypothesized that arboreal and terrestrial will both have a negative relationship between body mass and litter size but the strength of the relationship will differ. The hypothesis is partially supported because terrestrial species have a negative relationship consistent with r/K selection theory while arboreal species showed a positive relationship after PGLS which suggests arboreal species do not follow r/K selection theory. Based on row three of the PGLS output above, Terrestrial vs Arboreal, the estimate shows that terrestrial species have 2.88 more offspring than arboreal species at the same body size. Row two, the effect of body mass in arboreals, shows that arboreal species have a positive relationship, 0.56, meaning after accounting for phylogeny a bigger body mass means a larger litter size. Looking at the raw data on the scatterplot, arboreal species appear to have a negative relationship between body mass and litter size which is misleading because it reflects shared evolutionary history among the species. After PGLS, we are able to see the true relationship between body mass and litter size looking at the species independent of their evolutionary relationship. Row four, the difference in r/K slope shows that the r/K tradeoff is -1.0045 units more steep in terrestrial than in arboreal. This difference in the slopes directly addresses the hypothesis by suggesting that arboreal and terrestrial species have different relationship between body mass and litter size and that terrestrial environment enforce the r/K selection tradeoff while arboreal environments weaken it.

Conclusions & Acknowledgements

Our results are consistent with our expectations for arboreal and terrestrial natural history.

- Terrestrial environments have more predation and instability which explains why terrestrial species would have more r selected strategies, bigger litter sizes overall to increase chances of survival.
- Arboreal environments are likely more stable and have less predation and more resources which explains why the r/K tradeoff is weakened because arboreal species do not have the same ecological pressure meaning body mass is no longer a strong predictor of litter size reflected by the shallower slope.